Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 1 of 7 · Question 2 part c.i · 2 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part c.i in this mark scheme extract

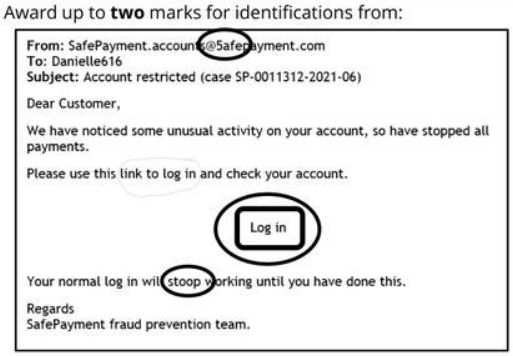
2(a)	**The only correct answer is B** *A is not correct because a password is needed to prevent unauthorised access to the network* *C is not correct because it only provides security after access to the network* *D is not correct because it only provides security after access to the network*		1

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	Award **one** mark from: • the email should be addressed to Danielle by her (full) name (1) • the email should be personalised (1)		1

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 1 of 7 · Question 2 part c.i · 2 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part c.i in this mark scheme extract

2(b)(ii)	Award up to **two** marks for identifications from:  **From:** SafePayment.account@5afepayment.com **To:** Danielle616 **Subject:** Account restricted (case SP-0011312-2021-06) Dear Customer, We have noticed some unusual activity on your account, so have stopped all payments. Please use this link to log in and check your account. Log in Your normal log in will stoop working until you have done this. Regards SafePayment fraud prevention team.	Allow link to log in OR circle around button	2

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award up to **two** marks for a linked description such as: A hacker/third party spies on/watches the user (of an electronic device) (1) In order to obtain their personal identification number/password/login information/sensitive information (1)		2

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 1 of 7 · Question 2 part c.i · 2 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part c.i in this mark scheme extract

2(c)(ii)	Award up to **two** marks for a linked explanation such as: • tilt the screen away from possible viewers/position yourself with your back to a wall (1) to ensure no one can see the screen (1) • shield your screen/keypad/keyboard when entering (sensitive/personal) information (1) to stop people seeing/memorising passwords/named sensitive item/sensitive/personal information (1) • use long/strong passwords (1) to prevent onlookers memorising them as you type (1) • use a screen/privacy filter (1) because it will prevent anyone not sitting directly in front of the screen from reading what is displayed (1) • sit where the information displayed on the screen (1) can't be captured on CCTV/by a drone-mounted camera or viewed by someone using binoculars (1)	Allow 1 mark for don't enter private information in a public place	2

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	**The only correct answer is B** *A is not correct because this is not the purpose of penetration testing* *C is not correct because this is not the purpose of penetration testing* *D is not correct because this is not the purpose of penetration testing*		1
2(d)(ii)	Award up to **two** marks for a linked description such as: • compromised/unpatched software is more vulnerable to attack (1) and may allow an attacker control of the whole network (1) • unpatched software has known weaknesses (1) which can be exploited by a hacker (1)		2
		Total	**11**

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 1 of 7 · Question 2 part c.i · 2 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 2 of 7 · Question 2 part b.i · 1 mark

2021 NOV · Paper 01 · extract 1 of 4 · find part b.i in this mark scheme extract

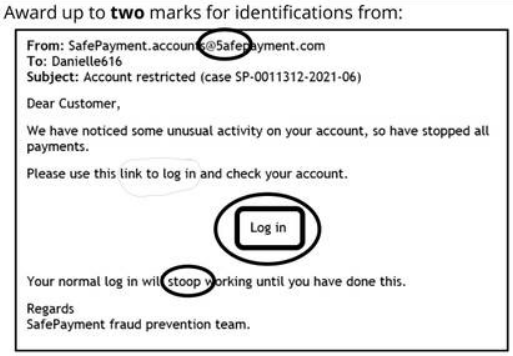
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Question Number	Answer	Additional Guidance	Mark
2(b)(i)	Award **one** mark from: • the email should be addressed to Danielle by her (full) name (1) • the email should be personalised (1)		1

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 2 of 7 · Question 2 part b.i · 1 mark

2021 NOV · Paper 01 · extract 2 of 4 · find part b.i in this mark scheme extract

2(b)(ii)	Award up to **two** marks for identifications from:  **From:** SafePayment.account@5afepayment.com **To:** Danielle616 **Subject:** Account restricted (case SP-0011312-2021-06) Dear Customer, We have noticed some unusual activity on your account, so have stopped all payments. Please use this link to log in and check your account. Log in Your normal log in will stoop working until you have done this. Regards SafePayment fraud prevention team.	Allow link to log in OR circle around button	2

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award up to **two** marks for a linked description such as: A hacker/third party spies on/watches the user (of an electronic device) (1) In order to obtain their personal identification number/password/login information/sensitive information (1)		2

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 2 of 7 · Question 2 part b.i · 1 mark

2021 NOV · Paper 01 · extract 3 of 4 · find part b.i in this mark scheme extract

2(c)(ii)	Award up to **two** marks for a linked explanation such as: • tilt the screen away from possible viewers/position yourself with your back to a wall (1) to ensure no one can see the screen (1) • shield your screen/keypad/keyboard when entering (sensitive/personal) information (1) to stop people seeing/memorising passwords/named sensitive item/sensitive/personal information (1) • use long/strong passwords (1) to prevent onlookers memorising them as you type (1) • use a screen/privacy filter (1) because it will prevent anyone not sitting directly in front of the screen from reading what is displayed (1) • sit where the information displayed on the screen (1) can't be captured on CCTV/by a drone-mounted camera or viewed by someone using binoculars (1)	Allow 1 mark for don't enter private information in a public place	2

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	**The only correct answer is B** *A is not correct because this is not the purpose of penetration testing* *C is not correct because this is not the purpose of penetration testing* *D is not correct because this is not the purpose of penetration testing*		1
2(d)(ii)	Award up to **two** marks for a linked description such as: • compromised/unpatched software is more vulnerable to attack (1) and may allow an attacker control of the whole network (1) • unpatched software has known weaknesses (1) which can be exploited by a hacker (1)		2
		Total	**11**

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 2 of 7 · Question 2 part b.i · 1 mark

2021 NOV · Paper 01 · extract 4 of 4 · find part b.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 3 of 7 · Question 2 part b.ii · 2 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

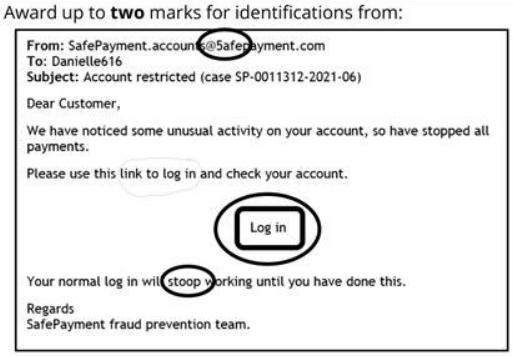
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Question Number	Answer	Additional Guidance	Mark
2(b)(i)	Award **one** mark from: • the email should be addressed to Danielle by her (full) name (1) • the email should be personalised (1)		1

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

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2021 NOV · Paper 01 · extract 2 of 4 · find part b.ii in this mark scheme extract

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Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award up to **two** marks for a linked description such as: A hacker/third party spies on/watches the user (of an electronic device) (1) In order to obtain their personal identification number/password/login information/sensitive information (1)		2

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 3 of 7 · Question 2 part b.ii · 2 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part b.ii in this mark scheme extract

2(c)(ii)	Award up to **two** marks for a linked explanation such as: • tilt the screen away from possible viewers/position yourself with your back to a wall (1) to ensure no one can see the screen (1) • shield your screen/keypad/keyboard when entering (sensitive/personal) information (1) to stop people seeing/memorising passwords/named sensitive item/sensitive/personal information (1) • use long/strong passwords (1) to prevent onlookers memorising them as you type (1) • use a screen/privacy filter (1) because it will prevent anyone not sitting directly in front of the screen from reading what is displayed (1) • sit where the information displayed on the screen (1) can't be captured on CCTV/by a drone-mounted camera or viewed by someone using binoculars (1)	Allow 1 mark for don't enter private information in a public place	2

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	**The only correct answer is B** *A is not correct because this is not the purpose of penetration testing* *C is not correct because this is not the purpose of penetration testing* *D is not correct because this is not the purpose of penetration testing*		1
2(d)(ii)	Award up to **two** marks for a linked description such as: • compromised/unpatched software is more vulnerable to attack (1) and may allow an attacker control of the whole network (1) • unpatched software has known weaknesses (1) which can be exploited by a hacker (1)		2
		Total	**11**

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 3 of 7 · Question 2 part b.ii · 2 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part b.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 4 of 7 · Question 2 part c.ii · 2 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

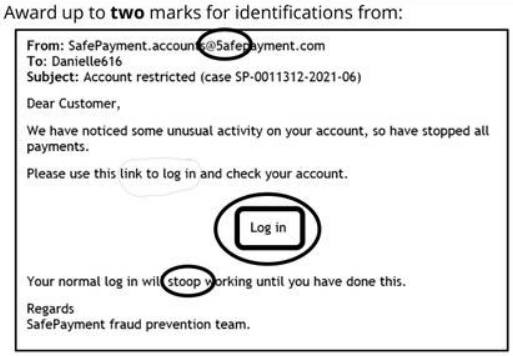
2(a)	**The only correct answer is B** *A is not correct because a password is needed to prevent unauthorised access to the network* *C is not correct because it only provides security after access to the network* *D is not correct because it only provides security after access to the network*		1

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	Award **one** mark from: • the email should be addressed to Danielle by her (full) name (1) • the email should be personalised (1)		1

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 4 of 7 · Question 2 part c.ii · 2 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

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Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award up to **two** marks for a linked description such as: A hacker/third party spies on/watches the user (of an electronic device) (1) In order to obtain their personal identification number/password/login information/sensitive information (1)		2

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 4 of 7 · Question 2 part c.ii · 2 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract

2(c)(ii)	Award up to **two** marks for a linked explanation such as: • tilt the screen away from possible viewers/position yourself with your back to a wall (1) to ensure no one can see the screen (1) • shield your screen/keypad/keyboard when entering (sensitive/personal) information (1) to stop people seeing/memorising passwords/named sensitive item/sensitive/personal information (1) • use long/strong passwords (1) to prevent onlookers memorising them as you type (1) • use a screen/privacy filter (1) because it will prevent anyone not sitting directly in front of the screen from reading what is displayed (1) • sit where the information displayed on the screen (1) can't be captured on CCTV/by a drone-mounted camera or viewed by someone using binoculars (1)	Allow 1 mark for don't enter private information in a public place	2

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	**The only correct answer is B** *A is not correct because this is not the purpose of penetration testing* *C is not correct because this is not the purpose of penetration testing* *D is not correct because this is not the purpose of penetration testing*		1
2(d)(ii)	Award up to **two** marks for a linked description such as: • compromised/unpatched software is more vulnerable to attack (1) and may allow an attacker control of the whole network (1) • unpatched software has known weaknesses (1) which can be exploited by a hacker (1)		2
		Total	**11**

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 4 of 7 · Question 2 part c.ii · 2 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 5 of 7 · Question 2 part d.ii · 2 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part d.ii in this mark scheme extract

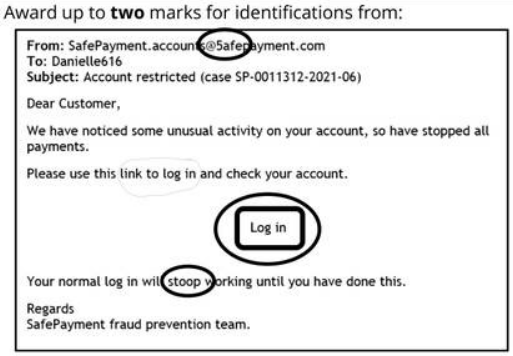
2(a)	**The only correct answer is B** *A is not correct because a password is needed to prevent unauthorised access to the network* *C is not correct because it only provides security after access to the network* *D is not correct because it only provides security after access to the network*		1

Question Number	Answer	Additional Guidance	Mark
2(b)(i)	Award **one** mark from: • the email should be addressed to Danielle by her (full) name (1) • the email should be personalised (1)		1

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 5 of 7 · Question 2 part d.ii · 2 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part d.ii in this mark scheme extract

2(b)(ii)	Award up to **two** marks for identifications from:  	Allow link to log in OR circle around button	2

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Award up to **two** marks for a linked description such as: A hacker/third party spies on/watches the user (of an electronic device) (1) In order to obtain their personal identification number/password/login information/sensitive information (1)		2

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 5 of 7 · Question 2 part d.ii · 2 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part d.ii in this mark scheme extract

2(c)(ii)	Award up to **two** marks for a linked explanation such as: • tilt the screen away from possible viewers/position yourself with your back to a wall (1) to ensure no one can see the screen (1) • shield your screen/keypad/keyboard when entering (sensitive/personal) information (1) to stop people seeing/memorising passwords/named sensitive item/sensitive/personal information (1) • use long/strong passwords (1) to prevent onlookers memorising them as you type (1) • use a screen/privacy filter (1) because it will prevent anyone not sitting directly in front of the screen from reading what is displayed (1) • sit where the information displayed on the screen (1) can't be captured on CCTV/by a drone-mounted camera or viewed by someone using binoculars (1)	Allow 1 mark for don't enter private information in a public place	2

Question Number	Answer	Additional Guidance	Mark
2(d)(i)	**The only correct answer is B** *A is not correct because this is not the purpose of penetration testing* *C is not correct because this is not the purpose of penetration testing* *D is not correct because this is not the purpose of penetration testing*		1
2(d)(ii)	Award up to **two** marks for a linked description such as: • compromised/unpatched software is more vulnerable to attack (1) and may allow an attacker control of the whole network (1) • unpatched software has known weaknesses (1) which can be exploited by a hacker (1)		2
		Total	**11**

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 5 of 7 · Question 2 part d.ii · 2 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part d.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 6 of 7 · Question 5 part c · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

5(a)(i)	Any **two** from: • It will interpret/analyse patient input to identify symptoms (1) and match the symptoms to (possible) illnesses (1) • It will match symptoms to possible illnesses (1) and give the most likely/probable illness (1) • It will match symptoms to possible illnesses (1) and ask further questions to narrow it down (1) • It will match symptoms to possible illnesses (1) by searching/using a database/other data store (1)		2
5(a)(ii)	Award **one** mark from: 1. May not have access to the internet (1) 2. May not have access to a device (1) 3. May not want to use it (1) 4. May not have the technical knowledge to use it (1) 5. May have a physical disability that stops them from using the service (1) 6. May not want to disclose personal information (1) 7. hacking or security or data issue (1) 8. May not trust the AI/system (1) 9. Can't take physical measurements eg. blood pressure (1) 10. may want a real person (1)		1

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 6 of 7 · Question 5 part c · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **one** mark for: • Local area network / LAN /VLAN		1
5(b)(ii)	Award **one** mark for: • Wide area network / WAN		1

Question Number	Answer	Additional Guidance	Mark
5(c)	**The only correct answer is C** *A is not correct because this is phishing* *B is not correct because this is shoulder surfing* *D is not correct because this is pharming*		1

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 6 of 7 · Question 5 part c · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award up to two marks for a linked explanation such as:
Robust/less likely to break it if dropped / quieter than a mechanical hard drive (1) because it doesn't have any moving/mechanical parts (1)
Smaller/thinner/lighter than a device with HDD (1) so easier to carry/more portable / laptops have less room in them for drives/componants(1)
SSD uses less power (1) so battery will last longer (1)
SSD is faster (1) so can access data/records more quickly (1)</td><td></td><td>2</td></tr>
<tr><td>5(d)(ii)</td><td>Any two from:
Flash memory chips are used (1)
Chips have (floating gate) transistors/electron pools/charge traps (1)
Gates/electron pools/charge traps hold an electrical charge (1)
Charge remains even when no power (1)
No charge/empty pool/trap represents data/1 (1)
A charge/full pool/trap represents no data/0 (1)
Data is stored in blocks and pages (1)</td><td></td><td>2</td></tr>
<tr><td>5(d)(iii)</td><td>Award one mark for each of:<table><tr><th>Description</th><th>RAM</th><th>ROM</th></tr><tr><td>Stores the boot up sequence</td><td></td><td>▯</td></tr><tr><td>The contents are lost when the laptop is shut down</td><td>▯</td><td></td></tr></table></td><td></td><td>2</td></tr>
<tr><td colspan="3">Total for question 5</td><td>12</td></tr>
</table>

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 6 of 7 · Question 5 part c · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 7 of 7 · Question 1 part b.iii · 1 mark

2023 JUN · Paper 01 · extract 1 of 4 · find part b.iii in this mark scheme extract

1(a)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Access (shared) peripherals (printers) (1) 2. Access the Internet (1) 3. Access shared file storage (server, cloud) (1) 4. Allows data/files/information to be shared (1) 5. Provide communication (email, social media, online meeting, chat) (1) 6. Provide collaboration (editing online documents, playing games) (1) 7. Allow hot-desking (1) 8. Allows centralised software updates/backups (1)	Award any example that can be categorised to a unique reason	2

Question Number	Answer	Additional Guidance	Mark
1(b)(i)	**The only correct answer is C** ***A*** *is not correct because compression does not encrypt a file* ***B*** *is not correct because* encryption does not ensure both sides use the same key ***D*** *is not correct because* encryption does not stop hackers accessing a network		1

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 7 of 7 · Question 1 part b.iii · 1 mark

2023 JUN · Paper 01 · extract 2 of 4 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(b)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. (Electronic/combination) lock on doors (1) 2. Swipe/key cards/biometrics (1) 3. Security guards/security patrol (1) 4. CCTV cameras / surveillance technology (1) 5. Asset tagging/marking (1)		2

Question Number	Answer	Additional Guidance	Mark
1(b)(iii)	Award **one** mark for any of the following: 1. It exploits/relies on human behaviours/tricks people (to reveal sensitive information) (1)	Do not award examples e.g. Pharming without further information.	1

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 7 of 7 · Question 1 part b.iii · 1 mark

2023 JUN · Paper 01 · extract 3 of 4 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(c)	**The only correct answer is D** ***A*** *is not correct because 5G does not have a* **high transmission latency** ***B*** *is not correct because a 5G does not have* ***built-in security*** ***C*** *is not correct because* 5G does not have a **lower bandwidth than 3G and 4G**		1

Question Number	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for: • Testing each block of code/subprogram/function independently (as it is written/finished) (1)	Do not accept module in isolation as this is in the question.	1

Type MS · Cyber attacks: phishing, shoulder surfing, eavesdropping and unpatched software

Answer 7 of 7 · Question 1 part b.iii · 1 mark

2023 JUN · Paper 01 · extract 4 of 4 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. Belong to/have membership in a professional society (1) 2. Attend computer science related conferences/gatherings (1) 3. Attaining/gaining training/educational opportunities (1) 4. Behave in ethical/legal/moral ways (1) 5. Stay up to date with changes in the field/read up-to-date publications (1) 6. Use responsible programming practices e.g. due diligence / testing / commenting code for maintainability (1) 7. Avoid bias when making design choices (1)	Award examples if they can be categorised to a unique bullet point Answers must focus on computer science rather than being generic Do not accept things that would be expected as part of an employee's role e.g. writing code	2

Question Number	Answer	Additional Guidance	Mark